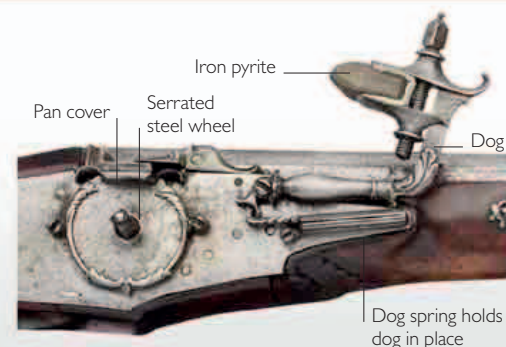


TURNING POINT

READY-TO-FIRE GUNS

Before 1500, all firearms had to be fired using a piece of smoldering match-cord. The device to hold this match-cord—the matchlock—was vulnerable to the effects of wind and rain, and the match-cord could potentially burn the user. The wheel-lock was the first mechanism to provide an internal system for igniting a firearm, allowing guns to be carried loaded and ready to fire in an instant. It made the development of an entirely new weapon—the pistol—possible, and revolutionized the use of firearms by cavalry.



▲ WHEEL-LOCK MECHANISM

A spring-loaded steel wheel sits under a pan. A piece of iron pyrite is held in jaws on a spring-loaded arm called a dog. Before firing, the dog is placed onto the pan cover. Pulling the trigger causes the wheel to spin as the pan cover opens, bringing the iron pyrite in contact with the wheel.

From their first appearance in Europe in the 14th century, firearms had to be lit and fired with the help of direct heat. The only practical source of this heat was hemp- or match-cord, impregnated with saltpeter, or potassium nitrate, which smoldered when lit. Early handguns were fired by match-cord held in the hand, which made supporting and aiming the gun difficult. Matchlocks were then devised to help place the lit match-cord into a priming pan. Burning match-cord, however, posed a constant risk to the shooter. Plus, it could be extinguished in bad weather.

» BEFORE

The match-cord and priming powder of matchlock guns could be rendered damp and useless in windy or rainy weather. The smoldering match-cord was also a source of danger to its user.

● **LARGE QUANTITIES OF MATCH-CORD** had to be supplied to armies since soldiers had to keep it burning in readiness, even if no gun was fired.

● **THE MATCH-CORD POSED A RISK** to a soldier because he kept it smoldering if his musket was likely to be fired. The match-cord could either burn him or set off his supply of gunpowder.

● **IMPOSSIBLE TO CONCEAL**, a matchlock weapon with a smoldering match-cord would easily give away the soldier's position at night.

● **ON HORSEBACK, IT WAS UNWIELDY AND IMPRACTICAL** to load and fire a matchlock weapon, and so cavalry, other than dragoons (mounted infantry), were not equipped with firearms.



MATCHLOCK MUSKET

